

Complete Build Roadmap

Eighteen days, two people, built from nothing. Every task states what to do, how to do it, and why it matters — with the full technology stack and the reasoning behind each choice.

PROJECT	Adaptive multi-persona AI interview panel
DAY 1	Thursday 20 August 2026 — today
SUBMISSION	Sunday 6 September 2026, 11:00 PM IST
GRAND FINALE	Saturday 12 September 2026, Delhi
TEAM	2 people — Person A (platform), Person B (intelligence)
BUDGET	Free tiers only, plus 300 free Agora agent-minutes

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2. Technology Stack

Every platform and library, what it does, why it was chosen over the alternatives, and what it costs.

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Role split, the single integration contract, the dependency chain, and the daily rhythm.

4. Week 1 — Foundation

Days 1–6. Scaffold, gateway, first spoken exchange, mock harness, the panel joins.

5. Week 2 — Deck and Intelligence

Days 7–13. Round 2 submission, tool calling, handoffs, analysis, difficulty, contradictions.

6. Week 3 — Finish and Submit

Days 14–18. Report, human-in-the-loop, integration freeze, assets, submission.

7. Budget and Risk

Agent-minute tracking, rate-limit strategy, and what to cut if time runs short.

1 What You Are Building

A candidate joins a voice call and is interviewed by three AI interviewers with different voices, different concerns, and a shared memory of everything the candidate has said. The panel adapts its difficulty, catches vague and contradictory answers, and produces a written assessment where every judgement cites a timestamped quote from the transcript.

1.1 The product in one page

A candidate opens a web page and joins an audio call. Three AI interviewers are already in the channel:

PERSONA	ROLE	WHAT THEY CARE ABOUT, AND WHEN THEY INTERRUPT
Priya	Technical interviewer	Correctness, depth, trade-offs. Interrupts when the candidate hand-waves over a mechanism. Accepts sound implementations without probing commercial impact — which is exactly what triggers a handoff to Arjun.
Arjun	Product manager	User impact, prioritisation, the cost of being wrong. Challenges technically correct answers that never mention a customer.
Meera	Behavioural interviewer	Collaboration, conflict, ownership. Runs role-play scenarios and pins candidates down when answers drift into generalities.

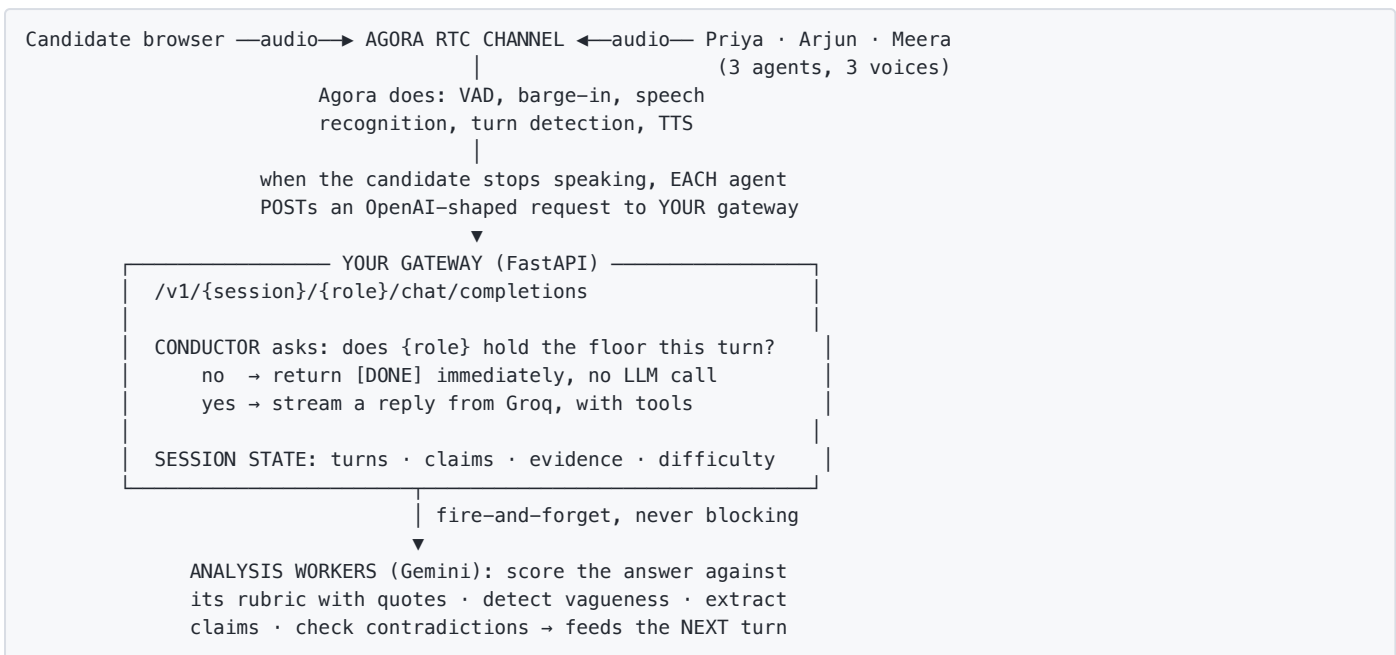
Only one speaks at a time. The candidate can talk over any of them and they stop mid-word. When the interview ends, a report renders with per-competency scores where every claim links to a specific turn in the transcript.

1.2 The eleven capabilities you are graded on

PS11 lists these explicitly. Treat it as a scoring rubric and build one demonstrable moment for each.

#	CAPABILITY	HOW THIS BUILD SATISFIES IT	DAY
1	Real-time, interruptible voice	Agora Conversational AI handles VAD and barge-in natively	3
2	Multiple interviewer roles	Three Agora agents, three TTS voices, three system prompts	6
3	Shared context between roles	A single <code>SessionState</code> digest injected into every persona prompt	10
4	Dynamic follow-up questions	Persona sees the scored answer and its missing concepts before replying	11
5	Controlled turn-taking	Conductor holds one <code>floor_holder</code> ; others return silence	6, 9
6	Role-play scenarios	Scenario library with the floor locked to the scenario owner	14
7	Difficulty adjustment	EWMA over answer scores with hysteresis, fed into prompts	12
8	Vague / contradictory answers	Punting-phrase and specificity checks; topic-narrowed claim conflict detection	5, 13
9	Evidence-based feedback	Scorer returns verbatim quotes; every report line cites turn ID and timestamp	5, 14
10	Structured final assessment	Per-competency aggregation rendered to PDF	14
11	Clear AI disclosure	Spoken greeting plus a persistent UI banner	10

1.3 How the pieces fit together



THE SINGLE IDEA THE WHOLE ARCHITECTURE RESTS ON

Agora lets you point an agent's `llm.url` at **any endpoint that speaks OpenAI's `/chat/completions` protocol**. So instead of writing an app that calls a model, you write a service that *pretends to be* a model, and Agora drives it.

That single fact gives you everything: three agents can share one brain, the conductor can silence a persona simply by returning an empty response, and all your interview logic lives in ordinary Python you can test without any voice infrastructure running.

2 Technology Stack

Every component below has a free tier except Agora, which is mandated by the rules and metered per agent-minute. Nothing here requires a credit card except the Agora account itself.

2.1 Real-time voice layer

TECHNOLOGY	WHAT IT DOES FOR YOU	WHY THIS AND NOT SOMETHING ELSE
Agora Conversational AI Engine REST API	The entire real-time pipeline: audio transport, voice activity detection, speech recognition, barge-in handling , turn detection, LLM invocation, and speech synthesis.	Mandatory — the rules require it as a core component, and "Use of Agora Conversational AI" is a scored criterion. It is also genuinely the right tool: building interruptible turn-taking yourself is weeks of work.
Agora Web SDK <code>agora-rtc-sdk-ng</code>	Browser client — joins the channel, publishes the microphone, subscribes to the agents' audio.	The only supported way for a browser to join an Agora channel. Loaded from a script tag; no build step.
Agora Token Builder <code>agora-token-builder</code>	Mints short-lived RTC tokens server-side for the candidate and each agent.	Channels require token auth. Never mint tokens in the browser — the App Certificate must stay server-side.
Agora-managed ASR & TTS	Speech recognition and the three distinct interviewer voices.	Included in the agent-minute price. Bringing your own key costs the same, and using a vendor outside the managed list (ElevenLabs, say) costs Agora's rate <i>plus</i> that vendor's bill. Managed is both cheaper and simpler.

2.2 Language models

PROVIDER	MODEL & FREE LIMITS	USED FOR	WHY
Groq primary	<code>llama-3.3-70b-versatile</code> 30 req/min · ~1,000 req/day No card required	Every spoken interviewer utterance	Speed. Groq's inference runs at hundreds of tokens per second, and "quality of voice/conversational experience" is explicitly scored. A free model that answers in 400 ms beats a frontier model that takes two seconds. It is also OpenAI-compatible, so the gateway passes frames straight through with no translation.
Google Gemini analysis only	<code>2.5-flash</code> / <code>2.5-flash-lite</code> 250–1,000 req/day 250K tokens/min	Rubrics, scoring, claims, contradictions, report	A separate quota pool — deliberately a different provider so heavy analysis can never starve the voice path. Also supports <code>response_schema</code> , which forces valid JSON instead of hoping for it.
Cerebras failover 1	Free tier, comparable speed	Voice path when Groq returns 429	Independent quota, similar latency profile, OpenAI-compatible. A one-line base-URL swap.
OpenRouter failover 2	Models with the <code>:free</code> suffix	Last resort on the voice path	Broad model choice and yet another independent quota. Slower, but a slow answer beats a rate-limit error on stage.

WHY NOT THE OBVIOUS ALTERNATIVES

OpenAI / Anthropic / other paid APIs — excellent models, but they cost money and the brief is zero budget.

Local Ollama — free, but disqualified twice over. Agora calls your LLM endpoint *over the public internet*, so `localhost:11434` is unreachable by definition; and an 8B model on a laptop adds seconds of latency to a scored criterion. Keep it only as an offline last-resort backstop behind a tunnel.

2.3 Backend

TECHNOLOGY	ROLE	WHY
Python 3.11	Everything server-side	Best library support for both Agora tokens and every LLM SDK you need.
FastAPI	The gateway — the OpenAI-compatible endpoint Agora calls	Native async plus <code>StreamingResponse</code> , which is exactly what server-sent events need. Flask would require extra machinery for streaming.
Uvicorn	ASGI server	Standard FastAPI pairing; handles many concurrent streaming connections.
<code>openai</code> SDK	Client for Groq, Cerebras and OpenRouter	Not OpenAI the service — the <i>library</i> , pointed at other base URLs. All three providers implement the same protocol, so one client class covers your whole fallback chain.
<code>google-genai</code>	Gemini client for the analysis path	Gives you <code>response_schema</code> , which is what makes free-tier JSON reliable enough to score against.
<code>httpx</code>	Async HTTP to Agora's REST API	Async-native, so joining three agents happens concurrently rather than serially.
PyYAML	Loads <code>personas.yaml</code>	Personas as data, not code — you will retune prompts many times and should never redeploy to do it.
python-dotenv	Loads <code>.env</code>	Keeps eight secrets out of source control.
WeasyPrint	Renders the assessment HTML to PDF	Free, local, no document API. You already write the report as HTML for the browser view, so PDF is nearly free.

2.4 State, frontend, hosting

CHOICE	ROLE	WHY
In-process dataclasses + JSON export	Turn, claim and evidence ledgers; session state	An interview is a few hundred rows that live for twenty minutes. A database is complexity you would spend half a day on for zero scoring benefit. Export to JSON on session end so nothing is lost. <i>Add SQLite only if you want history across restarts.</i>
Vanilla HTML + JS	Candidate client and observer view	No build step, no framework, no bundler config. React would cost half a day of setup and win no points — judges score the conversation, not the component library.
Hugging Face Spaces Docker SDK	Hosts the gateway on public HTTPS	Free, gives a stable public URL, and does not force-sleep CPU spaces — unlike free Render, whose cold start would make a judge's first click time out.
Cloudflare Tunnel <code>cloudflared</code>	Public HTTPS to your laptop during development	Free with no time limit. Lets Agora reach code running locally, so you iterate without redeploying. Use it for dev; use Spaces for the judge-facing URL.
GitHub Pages / Vercel	Static hosting for the web client	Free, instant, and keeps the frontend independent of gateway deploys.
Git + GitHub	Source control and the submission artefact	Required at submission. Commit history is also your evidence that the project was built during the hackathon.

2.5 What it all costs

COMPONENT	FREE ALLOWANCE	WHAT HAPPENS AT THE LIMIT
Agora Conversational AI	300 agent-minutes/month, then \$0.10 each	The only real cost. Bills per <i>agent</i> , so three personas × 10 minutes = 30 minutes. See §7.
Agora ASR + TTS	Included above	No separate bill.
Groq	30 req/min, ~1,000/day	HTTP 429 → chain advances to Cerebras.
Gemini	250–1,000 req/day	Batch several turns into one analysis call.
Cerebras / OpenRouter	Independent free quotas	Failover only; rarely touched.
Hugging Face Spaces	Free CPU tier	Sufficient — the gateway is I/O-bound, not compute-bound.
Everything else	Free	—

3 How the Two of You Work

3.1 The role split

PERSON	OWNS THESE DIRECTORIES	RESPONSIBLE FOR
Person A Platform & voice	agora/ gateway/ mock/ web/	Agent lifecycle, RTC tokens, the OpenAI-compatible endpoint, SSE streaming, provider fallback, the mock harness, browser client, deployment.
Person B Intelligence	conductor/ state/ analysis/	Floor control, handoff policy, ledgers, rubrics, scoring, contradiction detection, difficulty, the final report.

Why split by system boundary and not by feature: if you split by feature — "you do difficulty, I do contradictions" — both people end up editing the gateway, the conductor and the UI, and you spend the fortnight resolving merge conflicts. Split by directory and you almost never touch the same file.

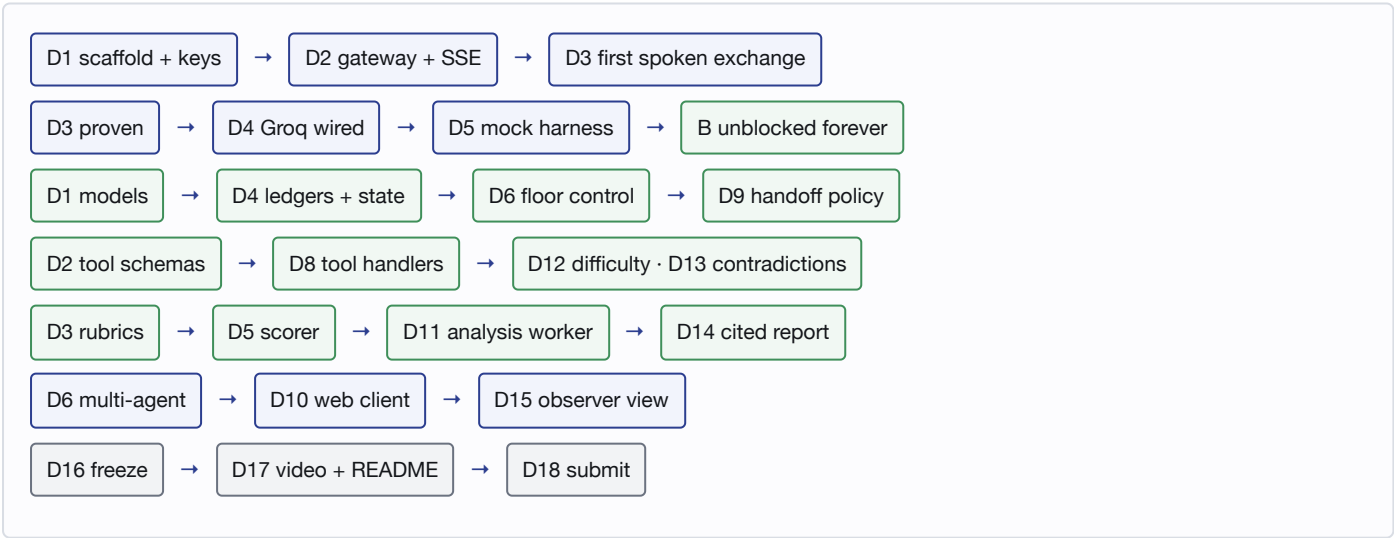
3.2 The one contract

WRITE THIS ON DAY 1 AND NEVER CHANGE IT

```
# src/contract.py – both people import from here
def next_utterance(session_id: str, role: str, messages: list) -> Iterator[str] | None:
    """None → this persona is silent this turn (gateway emits [DONE], no LLM call)
       iter → this persona holds the floor; gateway streams chunks to Agora"""
```

Why one function and not an API: it is the smallest possible surface between the two lanes. A builds all the HTTP, SSE, Agora and provider plumbing *around* it; B builds all the interview intelligence *behind* it. Neither needs to understand the other's half, and the mock harness calls the identical function — which is why B can work with Agora switched off.

3.3 The dependency chain



3.4 The daily rhythm

You work simultaneously, every day. There are only three hand-offs in the whole fortnight, and all three land at day boundaries:

WHEN	HAND-OFF	DOES IT BLOCK?
Day 1, first hour	A shares <code>.env</code> keys with B	Five minutes. B's only dependency all week.
End of Day 4	B's real <code>next_utterance()</code> → A wires the gateway to it	No — A uses a hardcoded reply until then.
End of Day 5	A's mock harness → B's permanent test loop	No — B unit-tests standalone until then.

After Day 5 there are no further hand-offs. Work in parallel through the day; merge for about thirty minutes each evening to run that day's checkpoint together.

WHY THE MOCK HARNESS MATTERS MORE THAN IT LOOKS

If Day 3's interrupt test fails and Person A spends three days on Discord debugging the platform, Person B still builds rubrics, ledgers, the scorer, floor control, tools, handoff policy, analysis, difficulty, contradictions and the report — all the way to Day 14 — **with Agora never once working.**

Only Person A's lane carries platform risk, and only Person A's lane spends agent-minutes. That is the entire reason the harness is scheduled on Day 5 rather than "whenever we get to it".

3.5 The file manifest

```

src/
├── contract.py           D1  the one shared signature
├── agora/
│   ├── tokens.py        D3  RTC token minting
│   ├── agent.py         D3  join / leave / speak / interrupt
│   └── session.py       D6  multi-agent lifecycle + teardown
├── gateway/
│   ├── app.py           D2  FastAPI, the OpenAI-compatible route
│   ├── sse.py           D2  SSE frame encoding
│   └── providers.py      D4  Groq · D9 Cerebras → OpenRouter fallback
├── conductor/
│   ├── personas.yaml    D2  three personas as data
│   ├── personas.py      D2  loader + prompt assembly
│   ├── tools.py         D2  schemas · D8 handlers
│   ├── conductor.py     D6  floor control · D9 handoff policy
│   ├── difficulty.py    D12 EWMA ladder
│   └── scenarios.py     D14 role-play library
├── state/
│   ├── models.py        D1  Turn, Claim, Evidence dataclasses
│   ├── ledger.py        D4  turn / claim / evidence ledgers
│   └── session.py       D4  SessionState
├── analysis/
│   ├── rubrics.py       D3  Gemini rubric generation
│   ├── scorer.py        D5  LLM judge + evidence quotes
│   ├── worker.py        D11 async post-turn analysis
│   ├── contradiction.py D13 topic-narrowed conflict check
│   └── report.py        D14 cited final assessment
├── mock/
│   └── replay.py         D5  scripted interview, 0 agent-minutes
└── web/
    ├── index.html       D10 candidate client
    └── observer.html    D15 recruiter view

```

4 Week 1 — Foundation (Days 1–6)

No user-facing features this week. The goal is a proven platform path and a mock harness, so that from Day 5 the two of you never wait on each other again. Resisting the urge to build interview logic now is the single most valuable discipline of the fortnight.

Day 1 — Thursday 20 August

Scaffold: accounts, keys, data models, and the contract between you

WHY TODAY MATTERS

Nothing can be built until both people can import the same package and A holds the credentials. This day looks like admin, but skipping any of it costs you a blocked afternoon later in the week.

PERSON A — ACCOUNTS AND ENVIRONMENT

Create the repository and Python environment

WHAT A GitHub repo with Python 3.11, a virtual environment, and pinned dependencies.

HOW

```
python -m venv .venv && . .venv/bin/activate
pip install fastapi uvicorn httpx openai google-genai \
    pyyaml python-dotenv agora-token-builder weasyprint
pip freeze > requirements.txt
```

WHY *Pinning today means Hugging Face Spaces builds the same environment on Day 2. Unpinned dependencies are the classic cause of "it works locally" on deployment day.*

Provision Agora

WHAT App ID, App Certificate, Customer ID, Customer Secret, and Conversational AI Engine enabled.

HOW Console → create project → copy App ID and App Certificate → RESTful API section for Customer ID and Secret → enable Conversational AI Engine on the project.

WHY *Two different credential pairs do two different jobs: the App Certificate signs RTC tokens, while Customer ID/Secret authenticate REST calls. Teams routinely confuse them and lose an hour on Day 3.*

Ask Discord about participant credits

WHAT A message asking whether credits exist beyond the 300 free agent-minutes.

HOW Post in the Agora support channel; name your team and problem statement.

WHY *The hackathon is run with Agora, and sponsors routinely issue credits that never appear on the public pricing page. This message has the best effort-to-payoff ratio of anything in the fortnight — it can remove your only hard constraint outright.*

Collect free LLM keys and write `.env`

WHAT Groq, Gemini and Cerebras keys, plus a shared secret you invent.

HOW

```
# .env — add to .gitignore BEFORE writing it
AGORA_APP_ID=                AGORA_APP_CERTIFICATE=
AGORA_CUSTOMER_ID=           AGORA_CUSTOMER_SECRET=
GROQ_API_KEY=                 GEMINI_API_KEY=
CEREBRAS_API_KEY=            GATEWAY_SHARED_SECRET=
PANEL_SIZE=3                 AGORA_IDLE_TIMEOUT=30
```

WHY `GATEWAY_SHARED_SECRET` is a value you choose — you put it in Agora's agent config as `llm.api_key` and verify it on every request. Without it your public endpoint is open to anyone who finds the URL.

Share the keys with Person B

WHAT The `.env` contents, sent privately.

WHY *This is B's only dependency on you all week. Doing it in the first hour means B is never blocked again.*

PERSON B — THE DOMAIN MODEL

Write the contract file `src/contract.py`

WHAT The `next_utterance()` signature and its docstring. No implementation.

WHY *It is the agreement between the two lanes. Writing it as real code rather than a chat message means neither of you can quietly drift from it.*

Write the data models `src/state/models.py`

WHAT Four dataclasses that everything else hangs off.

HOW

```
@dataclass
class Turn:
    turn_id: int; speaker: str; text: str
    started_at: float; ended_at: float
    difficulty: str

@dataclass
class Claim:          # for contradiction detection
    claim_id: int; turn_id: int
    text: str; topic: str; quote: str

@dataclass
class Evidence:       # for the cited report
    turn_id: int; competency: str
    quote: str; polarity: str
    concept: str; weight: float
```

WHY *Every one of these carries a `turn_id`, and that is deliberate. PS11 requires feedback "linked to the interview transcript" — if the identifier is present from the first line of code, citation is free later. Retrofitting it on Day 14 means rewriting the scorer, the ledgers and the report.*

Create the package tree

WHAT All folders from the manifest in §3.5, each with an `__init__.py`.

WHY *Empty packages make imports resolve from day one. Creating them as you go produces a week of `ModuleNotFoundError` during merges.*

Write a throwaway stub

WHAT `next_utterance()` returning `iter(["Hello, shall we begin?"])` for `"technical"` and `None` for everything else.

WHY *A needs something callable tomorrow to prove the gateway works. Ten lines now saves a blocked morning.*

EVENING CHECKPOINT Both of you can run `python -c "from src.contract import next_utterance"` · `.env` holds eight values · repo pushed

Agora 0 · total 0

Day 2 — Friday 21 August

A gateway that talks: public HTTPS, streaming, correct frame format

WHY TODAY MATTERS

Agora will only ever see one thing from you — an HTTP endpoint that streams OpenAI-shaped frames. Get the format wrong and the agent hangs or reads garbage aloud. Prove it works with a hardcoded reply *before* adding a model, so that when something breaks tomorrow you know it is not the framing.

PERSON A — THE ENDPOINT

Write the SSE encoder `src/gateway/sse.py`

WHAT A generator that turns text chunks into server-sent event frames.

HOW

```
def sse(chunks):
    for c in chunks:
        f = {"choices":[{"index":0,
            "delta":{"content": c}]}]}
        yield f"data: {json.dumps(f)}\n\n"
    yield "data: [DONE]\n\n"
```

WHY Two details are non-negotiable. The **blank line after every frame** is what makes it a valid SSE event — without it the client buffers forever. And **[DONE]** is mandatory: omit it and the agent waits for more output until its idle timeout, which looks exactly like a crash.

Write the route `src/gateway/app.py`

WHAT A FastAPI POST route that authenticates, calls `next_utterance`, and streams the result.

HOW

```
POST /v1/{session}/{role}/chat/completions
1. verify Authorization against GATEWAY_SHARED_SECRET
2. body = await req.json() # OpenAI shape
3. stream = next_utterance(session, role, body["messages"])
4. if stream is None: return sse([]) # [DONE] only
5. return StreamingResponse(sse(stream),
    media_type="text/event-stream")
```

WHY The `role` lives in the URL rather than the body because Agora sends a fixed body per agent — the path is the only place you can distinguish which persona is calling. Step 4 is the silence mechanism the whole turn-taking design depends on: no model call, no cost, no speech.

Deploy to Hugging Face Spaces

WHAT A public HTTPS URL that streams.

HOW New Space → Docker SDK → Dockerfile running `uvicorn` on port 7860 → set secrets in Space settings, not in the repo.

WHY Test it with `curl` from a phone hotspot, not from your laptop. If it responds from a network you do not control, Agora's servers can reach it too. This is the single check that prevents a wasted Day 3.

PERSON B — THE PANEL'S VOICES

Define the personas as data `src/conductor/personas.yaml`

WHAT Three entries: role key, display name, Agora voice ID, question domains, handoff triggers, system prompt.

HOW

```
technical:
  name: Priya
  voice: <agora voice id>
  domains: [systems, data, tradeoffs]
  handoff_when: [no_business_framing]
  prompt: |
    You are Priya, a staff engineer...
  product: {name: Arjun, ...}
  behavioural: {name: Meera, ...}
```

WHY YAML rather than Python constants because you will retune these prompts perhaps twenty times, including on Day 15 when tuning is the highest-value work available. Config means editing a file; code means a redeploy each time.

Write the three system prompts

WHAT Under 150 words each, with a distinct concern, an explicit interrupt condition, and one example of the desired tone.

WHY Two reasons for the length cap. Every token is resent on every turn, so a 900-word prompt across 25 turns will hit Groq's per-minute token ceiling mid-interview. And smaller models follow three sharp instructions far better than fifteen vague ones.

Write the loader `src/conductor/personas.py`

WHAT `persona_prompt(role, difficulty, shared_context)` returning the assembled system message.

WHY Assembling the prompt in one place means Day 10's shared context and Day 12's difficulty level both plug into a single function instead of being scattered through the codebase.

Draft the tool schemas only `src/conductor/tools.py`

WHAT Six JSON schemas — `request_handoff`, `flag_vague_answer`, `flag_contradiction`, `adjust_difficulty`, `launch_scenario`, `record_evidence`. No handlers yet.

WHY *A needs the exact shapes on Day 8 to write the stream interceptor. Defining them now removes a mid-week negotiation. Use `strict: true` and `additionalProperties: false` so the model cannot invent fields.*

EVENING CHECKPOINT `curl` your public HTTPS URL from outside your own network and receive streamed SSE frames ending in `[DONE]`

Agora 0 · total 0

Day 3 — Saturday 22 August

First contact — the gate that decides whether this fortnight is calm or frantic

WHY TODAY MATTERS

Every remaining day assumes a candidate can hold an interruptible spoken conversation with an Agora agent driven by your gateway. If that assumption is wrong, you need to know today — with thirteen working days left and the Agora support team reachable — rather than in September.

PERSON A — TALK TO AN AGENT

Mint RTC tokens `src/agora/tokens.py`

WHAT `build_token(channel, uid)` using the App Certificate.

WHY *Agora channels require token auth, and both the candidate and every agent need one. This must be server-side — putting the App Certificate in browser JavaScript hands anyone your account.*

Write join and leave `src/agora/agent.py`

WHAT Two REST calls that start and stop an agent.

HOW

```
POST api.agora.io/api/conversational-ai-agent/
v2/projects/{APPID}/join
Authorization: Basic b64(CUST_ID:CUST_SECRET)

properties.channel      = "interview-s1"
properties.token        = <agent rtc token>
properties.agent_rtc_uid = "1001"
properties.remote_rtc_uids= ["<candidate uid>"]
properties.idle_timeout = 30
properties.llm.style     = "openai"
properties.llm.url       = https://<you>/v1/s1/
                        technical/chat/completions
properties.llm.api_key   = <GATEWAY_SHARED_SECRET>
properties.tts           = <managed voice>
```

WHY `remote_rtc_uids` controls which participants the agent listens to. Note it now — if empty completions turn out not to silence an agent, swapping this list is your fallback turn-taking mechanism.

Use Agora's sample web client, not your own

WHAT Join the channel from Agora's published demo page or basic sample.

WHY *Today you are testing whether the platform and your gateway work together. Writing your own client at the same time means that when nothing happens, you cannot tell which half is broken. Your client comes on Day 10.*

Hold a conversation, then interrupt it

WHAT Speak over the agent mid-sentence. It must stop and yield.

WHY *This is the gate. PS11 capability 1 is "real-time and interruptible", and it is the one thing you cannot build yourself in the time available. If it does not work, post on the Agora Discord today.*

Guarantee teardown from the very first test

WHAT `leave` wrapped in a `finally` block.

WHY *An agent left running after a crashed test keeps billing until its idle timeout. This is the most common way hackathon teams quietly lose their free minutes, and it is entirely preventable with three lines.*

PERSON B — THE RUBRIC ENGINE

Generate rubrics with Gemini `src/analysis/rubrics.py`

WHAT `generate_rubric(question, role)` returning five to seven weighted concepts summing to 10.

HOW

```
generate_rubric(question, role) -> dict
{"concepts": [{"concept": str,
               "weight": float}]}
# use Gemini response_schema to force shape
```

WHY A rubric is what turns "the model liked the answer" into a defensible score. Without one you cannot say which concept was missing, and PS11 capability 9 requires exactly that specificity.

Normalise the weights

WHAT Rescale by `10 / total`; if the total is zero, split evenly.

WHY Models routinely return weights summing to 9 or 12 despite being told otherwise. Without normalisation your 10-point scale silently becomes a 12-point scale on some questions, and scores stop being comparable across the interview.

Guard the JSON parse and log every fallback

WHAT Extract with `raw.find("{")` and `rfind("}")` before `json.loads`. On failure, use a default rubric **and log a warning**.

WHY Free-tier models sometimes wrap JSON in prose. A bare `except` that silently returns a generic two-concept rubric produces scores that look perfectly reasonable and mean nothing at all — and you will not discover it until the judges ask how scoring works. Loud logging turns a silent corruption into a visible bug.

Verify across all three domains

WHAT Three questions per persona domain; confirm 5–7 substantive concepts each.

WHY A rubric prompt tuned only on systems-design questions often produces vague filler for behavioural ones. Catching that now is far cheaper than on Day 11 when the scorer depends on it.

GATE You interrupted an AI mid-sentence and it stopped. **If not, post on the Agora Discord today** — do not carry **Agora ~10 · total 10** this into next week.

Day 4 — Sunday 23 August

A real model behind the gateway, and the silence assumption tested

WHY TODAY MATTERS

Your entire turn-taking design assumes that returning an empty completion makes an agent stay quiet. That assumption has never been tested. Testing it today, while there is time to build the alternative, is the difference between an inconvenience and a crisis.

PERSON A — GROQ AND SILENCE

Wire Groq into the gateway `src/gateway/providers.py`

WHAT Replace the hardcoded reply with real streamed model output.

HOW

```
client = OpenAI(
    base_url="https://api.groq.com/openai/v1",
    api_key=GROQ_API_KEY)

stream(system, messages, tools) →
    model="llama-3.3-70b-versatile"
    stream=True, max_tokens=300, temperature=0.6
```

WHY Groq implements OpenAI's protocol exactly, so this is a base-URL change rather than a rewrite — and it means the same client class covers Cerebras and OpenRouter on Day 9 too.

Cap output at 300 tokens

WHAT `max_tokens=300` on every utterance call.

WHY Two benefits at once. Real interviewers are terse, so a cap improves realism — a monologuing interviewer is the fastest way to make a demo feel fake. And short replies stretch the free-tier token-per-minute quota across a longer interview.

Validate the silent-agent path

WHAT Join a second agent whose route returns `[DONE]` immediately. Confirm it produces genuine silence.

WHY *If empty completions work, turn-taking is pure application logic you fully control and can explain on stage. If they produce an artefact or an error, you need the fallback below — and you need to know today, not on 2 September.*

If silence fails, switch to the fallback design

WHAT Join non-active agents with `remote_rtc_uids: []` and swap subscriptions via the `update` endpoint on each handoff.

WHY *An agent subscribed to nobody cannot hear the candidate and therefore cannot respond. More moving parts and a little handoff latency, which is why it is the fallback — but it works without relying on any undocumented behaviour.*

PERSON B — MEMORY

Build the three ledgers `src/state/ledger.py`

WHAT Turn, claim and evidence ledgers with append and query methods.

HOW

```
TurnLedger.add(speaker, text) -> turn_id
    # stamps started_at / ended_at
ClaimLedger.add(turn_id, claim, topic, quote)
ClaimLedger.by_topic(topic) # ← D13 needs this
EvidenceLedger.add(turn_id, competency, quote,
    polarity) # ← D14 needs this
```

WHY *Build these before any conductor logic, because three later capabilities read from them and none can be retrofitted cheaply. `by_topic()` in particular is what makes contradiction detection tractable on a small model — see Day 13.*

Build session state `src/state/session.py`

WHAT The object every persona and worker reads from.

HOW

```
SessionState:
    floor_holder, difficulty, ewma
    consecutive_turns, pending_handoff
    active_scenario, rubrics{}
    turns, claims, evidence
    shared_digest(last_n=8)
```

WHY *A single shared object is precisely what PS11 capability 3 asks for. Because all three personas read the same state, cross-role memory is a property of the design rather than a feature you have to build.*

Write the real `next_utterance()`

WHAT Replace the stub: check `floor_holder`, return `None` if this role does not hold it, otherwise return canned text.

WHY *A wires the gateway to this tonight. Canned text is deliberate — you are proving the routing, and adding model calls now would confuse two separate failure modes.*

EVENING CHECKPOINT A real Llama answer reaches you through Agora · you know whether empty completions produce silence · turns are recorded with timestamps

Agora ~10 · total 20

Day 5 — Monday 24 August

The mock harness — after today Person B never needs Agora again

WHY TODAY MATTERS

This is the highest-leverage day of the fortnight. A script that impersonates Agora lets Person B build and test the entire interview brain for free, forever. It protects your agent-minute budget, removes platform risk from B's lane, and turns a slow manual test into a one-minute command.

PERSON A — MOCK REPLAY

Write the harness `src/mock/replay.py`

- WHAT** A script that posts a scripted interview into the gateway in Agora's exact request shape, calling *every* persona per turn, and asserting exactly one speaks.
- WHY** *It must call every persona, not just the floor holder — that is what Agora does, and the "exactly one speaks" assertion is what catches a broken conductor immediately rather than during a live session.*

Seed the script with deliberate bait

WHAT Five scripted answers, each designed to trigger a specific capability.

HOW

```
1 "shard by tenant id, routing
  table in Redis"      → strong answer
2 "latency 400ms → 60ms" → records a claim
3 "actually nearer 200ms" → CONTRADICTION
4 "it depends, you align
  stakeholders"        → VAGUE
5 strong tech, zero mention
  of a customer         → HANDOFF trigger
```

WHY *Contradictions, vagueness and cross-role handoffs are the three hardest capabilities and the easiest to break silently. Baking them into the default test means every run exercises them, without a single agent-minute.*

Make it the default test loop

WHAT `python -m src.mock.replay` prints the full interview and passes its assertions in under a minute.

WHY *If it is slower than a minute, people stop running it. Speed is what makes a test habitual.*

PERSON B — THE SCORER

Write the evidence-producing judge `src/analysis/scorer.py`

WHAT Ask the model which rubric concepts the answer covered **and for the verbatim quote proving each**.

HOW

```
score_answer(question, answer, rubric) →
{"concepts":[{"concept":"sharding",
                  "covered": true,
                  "quote":"shard by tenant id"}],
 "depth":"moderate",
 "missing":["failure modes"]}

score = Σ weight(covered), capped at 10
```

WHY *Do not keyword-match. Substring matching on rubric concepts marks almost every concept as covered and inflates every score toward 10 — a failure that produces plausible numbers and is nearly invisible. Requiring a quote also solves PS11 capability 9 in the same call: the score and its citation are produced together.*

Add vagueness detection

WHAT Two signals: a punting-phrase list ("it depends", "generally", "various factors") and a specificity check — does the answer contain numbers, named tools, or a concrete example? Penalise up to two points.

WHY *Half of PS11 capability 8, and cheap. The phrase list catches deflection; the specificity check catches fluent answers that say nothing — which smaller models tend to reward if you only ask "is this a good answer?".*

Unit-test standalone

WHAT One fixed rubric plus three hand-written answers: strong, vague, wrong.

WHY *You do not need the harness for this, so you are not blocked waiting for A. Three fixed cases also give you a regression check for when you tune the prompt later.*

EVENING CHECKPOINT `python -m src.mock.replay` drives a complete 10-turn interview end to end with **zero** **Agora 0 · total 20** agent-minutes consumed

Day 6 — Tuesday 25 August

Three voices in one channel — and registration closes today

WHY TODAY MATTERS

This is where it stops being a chatbot and becomes a panel. It is also the last day to register the team, and no amount of engineering compensates for missing that.

PERSON A — THE PANEL JOINS

Confirm Commudle registration — do this first

WHAT Both members registered, team of two to four.

WHY *Registration closes today and there is no appeal. Solo entries are not eligible regardless of project quality.*

Multi-agent session lifecycle `src/agora/session.py`

WHAT Join `PANEL_SIZE` agents with distinct UUIDs, voices and role segments; stop them all together.

HOW

```
start(channel):
    for role in personas[:PANEL_SIZE]:
        join(uuid, voice, f".../{role}/chat/completions")
stop():
    for aid in agents: leave(aid)
# both inside try/finally
```

WHY *Each agent needs its own URL segment so the gateway can tell which persona is asking. Teardown must cover all agents — one survivor drains the budget silently for as long as it takes to notice.*

Make `PANEL_SIZE` an environment variable

WHAT Read from `.env`, not a constant.

WHY *It is your credit-pressure release valve. Three personas cost 30 agent-minutes per ten-minute interview; two cost 20. If credits get tight in the final week you change one value instead of touching code — and PS11 says "one or more AI interviewers", so two is fully compliant.*

PERSON B — FLOOR CONTROL

Write the conductor `src/conductor/conductor.py`

WHAT One `floor_holder` per turn; holder streams, everyone else returns `None`.

HOW

```
holds_floor(session, role) -> bool
decide_floor(session, analysis) -> role
# round-robin today; real policy on Day 9
```

WHY *Three agents all hear the candidate, so all three call your gateway. The conductor is the only thing preventing them from answering simultaneously — and because the silent path performs no inference, two-thirds of your model calls cost nothing.*

Add a permanent one-speaker assertion

WHAT The harness asserts exactly one persona returns a stream per turn.

WHY *Overlapping voices is the most embarrassing possible demo failure and the easiest to introduce accidentally when editing handoff rules on Day 9. This assertion protects you for the rest of the project.*

Pre-generate rubrics at session start

WHAT One rubric per persona domain, stored in `SessionState.rubrics`.

WHY *Generating a rubric takes seconds. Doing it inside the response path would add that delay to a live conversation, which is directly scored. Pre-generation moves the cost to session setup where nobody notices.*

EVENING CHECKPOINT Registration confirmed · three distinct voices in one channel · exactly one speaks per turn · **Agora ~15 · total 35**
you can interrupt any of them (*four-minute test sessions only*)

5 Week 2 — Deck and Intelligence (Days 7–13)

Days 7 to 9 carry the Round 2 deck alongside the build — it is a shortlisting gate, so it cannot slip. From Day 10 the panel is real and the rest of the week is about making it think.

Day 7 — Wednesday 26 August

Deck day — both people, build paused

WHY TODAY MATTERS

Round 2 shortlists the top 250–500 teams. A brilliant prototype that never gets shortlisted scores nothing, so one full day on the deck is a rational trade even though it feels like lost build time.

PERSON A — THE ARCHITECTURE HALF

System diagram and Agora integration slide

WHAT

One diagram of the channel, gateway and workers; one slide naming the exact endpoints you use — `join` , `speak` , `interrupt` , `update` , `history` .

WHY

"Use of Agora Conversational AI" is a scored criterion, and screeners reading hundreds of decks reward specificity. Naming endpoints proves you have actually built against the platform rather than planning to.

Stack slide

WHAT

Agora + Groq + Gemini + FastAPI, stated plainly as running on free tiers.

WHY

Running a real-time multi-agent system at zero marginal cost is a genuine engineering result. It also speaks directly to the finale's scalability criterion.

PERSON B — THE STORY HALF

Name the product and lead with the panel

WHAT

A real name, and the multi-persona panel in the first two slides.

WHY

Most PS11 entries will be single-interviewer bots. The conducted panel is your differentiator, so it belongs where a screener will definitely see it — not in the appendix.

Frame the impact commercially

WHAT

Panel time is the scarcest resource in hiring; this replaces screening rounds rather than replacing quizzes.

WHY

Round 5 judges on business potential and real-world impact. Establishing that framing now means the finale deck is a refinement rather than a rewrite.

Map every PS11 capability to a feature

WHAT

One slide, eleven rows, each with how you address it.

WHY

Screeners are matching submissions against that list. Doing the matching for them removes any doubt about coverage.

EVENING CHECKPOINT

Deck drafted covering problem, solution, architecture, impact, and Agora integration

Agora 0 · total 35

Day 8 — Thursday 27 August

Tool calling — and the one bug that gets read aloud

WHY TODAY MATTERS

Tool calls are how the personas control the interview: handing over the floor, flagging a contradiction, raising the difficulty. The hackathon explicitly highlights "AI agents with tool calling", and here it does real work rather than being decorative.

PERSON A — INTERCEPT TOOLS IN THE STREAM

Strip tool deltas before forwarding to Agora

WHAT Forward only text content; route tool-call deltas to the conductor.

WHY *A leaked tool call is synthesised and read aloud as gibberish. The candidate hears the agent recite JSON. Test this deliberately today rather than discovering it in the demo video.*

Buffer partial tool JSON

WHAT Accumulate arguments per `tool_call.id` until the call is complete, then dispatch.

WHY *Tool arguments arrive in fragments across many chunks. Parsing each fragment individually throws on invalid JSON; you must reassemble first.*

PERSON B — THE SIX HANDLERS

Implement all six tools `src/conductor/tools.py`

HOW

```
request_handoff(to_role, reason)
    → session.pending_handoff
flag_vague_answer(quote, why)
flag_contradiction(turn_a, turn_b)
adjust_difficulty(direction)
launch_scenario(scenario_id)
record_evidence(competency, quote)
    → EvidenceLedger
```

WHY *These six tools implement five PS11 capabilities through one mechanism — turn-taking, vagueness, contradictions, difficulty and role-play. They are also visible on stage: surfacing tool calls in the UI on Day 12 is the cheapest way to make invisible reasoning legible to a judge.*

Always parse with `json.loads`

WHY *Models vary their string escaping, so substring matching on serialised arguments breaks unpredictably. Parse properly and the variation stops mattering.*

Test each tool through the harness

WHAT Force each tool from a scripted answer and assert the state change landed.

WHY *A tool that fires but writes nowhere looks identical to a working one in the logs, right up until the report comes out empty on Day 14.*

EVENING CHECKPOINT A tool call fires, changes conductor state, and is **not** spoken aloud

Agora ~5 · total 40

Day 9 — Friday 28 August

Handoff policy — and the deck is due today

WHY TODAY MATTERS

Today the panel stops rotating mechanically and starts handing over *for reasons*. That difference is what separates a demo that looks scripted from one that looks intelligent.

PERSON A — RESILIENCE AND SUBMISSION

Build the provider fallback chain

WHAT Catch HTTP 429 and advance: Groq → Cerebras → OpenRouter.

WHY *A paid API degrades under load; a free tier returns 429 and stops. On stage that is indistinguishable from a crash. Since all three providers speak the same protocol, the chain is a base-URL and key swap — very cheap insurance against your most likely live failure.*

Log which provider served each turn

WHY *When a judge asks why one answer took longer, you want a log line rather than a guess.*

Submit the Round 2 deck — in the morning

WHY *Evening submissions lose to upload failures and portal timeouts. Submit early, then confirm it actually landed.*

PERSON B — WHO SPEAKS NEXT, AND WHY

Implement the handoff policy `conductor.decide_floor()`

HOW

pending_handoff	→ the named role
contradiction	→ the original asker
tech ok + no biz	→ product
3 turns held	→ rotate
2 vague in a row	→ hold, pin down
scenario active	→ locked to owner

WHY Each rule exists for a reason. Routing contradictions to the original asker is natural because they have standing to challenge. The three-turn cap keeps all personas visibly active, so a judge sees a panel rather than one bot with two silent extras.

Make the PS11 scenario rule fire reliably

WHAT Technical accepts a sound answer that never mentions a customer → floor moves to Product.

WHY This is the exact example in the problem statement, and it is your demo centrepiece. Reproducing the organisers' own scenario signals that you read the brief carefully — and it demonstrates capabilities 2, 3 and 5 in a single twenty-second moment.

EVENING CHECKPOINT Round 2 deck submitted · the mock run shows Priya accepting and Arjun challenging, with a **Agora 0 · total 40** stated reason

Day 10 — Saturday 29 August

The panel works — your first real multi-persona interview

WHY TODAY MATTERS

Everything built so far comes together into something a person can actually experience. Record this session — it becomes your fallback footage if anything goes wrong on demo day.

PERSON A — THE CANDIDATE CLIENT

Build the web client `src/web/index.html`

WHAT Agora Web SDK: join channel, publish microphone, subscribe to all agents, render the live transcript.

WHY Plain HTML and vanilla JavaScript deliberately — no build step means no bundler configuration to debug at 2 a.m. on Day 17. Judges score the conversation, not the framework.

Add the AI disclosure

WHAT The lead persona's `greeting_message` states plainly that all interviewers are AI, plus a persistent UI banner.

WHY PS11 capability 11, complete, for about fifteen minutes of work. It is also the right thing to do — a candidate should never be uncertain whether they are talking to a person.

PERSON B — SHARED MEMORY

Implement the shared context digest `SessionState.shared_digest()`

WHAT Each persona's prompt receives a compact summary of what the candidate told the *other* personas.

WHY This is the mechanism behind PS11 capability 3, and it is what makes the panel feel like one interview rather than three separate ones. When Arjun references something the candidate told Priya, unprompted, that is the moment the concept lands for a judge.

Cap the context window hard

WHAT Last eight to ten turns plus a rolling summary. Set Agora's `max_history` to match.

WHY Unbounded history grows every turn and will hit Groq's tokens-per-minute ceiling partway through a long interview — failing exactly when the interview gets interesting. The rolling summary preserves early context at a fraction of the tokens.

MILESTONE A real three-persona spoken interview with a live handoff, an interruption, and cross-persona memory. **Agora ~25 · total 65** Record it.

Day 11 — Sunday 30 August

WHY TODAY MATTERS

The panel can talk; now it needs to understand. The critical constraint is that understanding must never slow the conversation down, because conversational quality is scored and a two-second pause after every answer destroys it.

PERSON A — MAKE THE PANEL LEGIBLE

Speaking indicator and attributed transcript

WHAT Panel avatars with an unmistakable "speaking now" state; transcript lines colour-coded per persona.

WHY *In a recorded demo, distinct voices alone are ambiguous — viewers with poor audio cannot tell the personas apart. A visual speaker indicator makes turn-taking obvious even on a laptop speaker in a noisy room.*

Switch daily testing to `PANEL_SIZE=2`

WHY *Saves a third of every test session's agent-minutes for the next five days. Switch back to three for Day 16 integration and the Day 17 video.*

PERSON B — THE ASYNC WORKER

Build the post-turn worker `src/analysis/worker.py`

WHAT Fire-and-forget after each candidate turn: score with evidence, vagueness check, claim extraction.

HOW

```
asyncio.create_task(analyse(turn))
# result lands in SessionState;
# the conductor reads it on the NEXT turn
```

WHY *Analysis takes one to three seconds. Waiting for it before replying would add that to every single turn. Feeding it into the next turn instead costs you nothing perceptible — the interviewer's follow-up is informed by the scored answer, which is exactly what PS11 capability 4 asks for.*

Extract claims per turn

WHAT A tiny schema: claim text, topic, and the verbatim quote.

WHY *The `topic` field is what makes Day 13's contradiction detection tractable. Keep the schema minimal — the more fields you ask a small model to fill, the less reliable each one becomes.*

Prove it is non-blocking

WHAT Measure response latency with the worker enabled and disabled.

WHY *An `await` in the wrong place silently converts your async worker into a synchronous one. The only reliable way to catch it is to measure.*

EVENING CHECKPOINT Analysis runs after every turn with no measurable latency cost · claims accumulate with quotes **Agora ~10 · total 75**

Day 12 — Monday 31 August

Adaptation — the interview responds to how the candidate is doing

WHY TODAY MATTERS

PS11 asks for difficulty that adapts to performance. This is the capability most often faked with a fixed question ladder, and the one a judge can test in thirty seconds by deliberately answering badly.

PERSON A — SHOW THE MACHINERY

Difficulty meter and live tool-call feed

WHAT A visible difficulty indicator, and tool calls surfaced as they fire.

WHY *Judges cannot score reasoning they cannot see. Everything the conductor does is invisible by nature — displaying it is the cheapest possible way to convert hidden intelligence into demo footage.*

PERSON B — THE DIFFICULTY LADDER

Write the controller `src/conductor/difficulty.py`

HOW

```
ewma = 0.4 * score + 0.6 * ewma
band: <4 easy | 4-7 medium | >7 hard
shift only after 2 consecutive
      out-of-band turns
expose level → persona prompt
              → UI meter
              → turn ledger
```

WHY *The exponentially weighted average smooths out a single unlucky answer, and the two-turn hysteresis stops the level oscillating between easy and hard on alternating turns — which looks broken rather than adaptive. Recording the level in the turn ledger also lets the final report say "scored 7 on a hard question", which is far more informative than a bare number.*

Verify both directions

WHAT Two weak scripted answers must lower the level; two strong ones must raise it.

WHY *Teams usually test only the downward path. A ladder that drops but never recovers punishes a candidate for one bad answer and is trivially exposed on stage.*

EVENING CHECKPOINT Two weak answers drop the difficulty level — audibly in the questions and visibly in the UI **Agora ~10 · total 85**

Day 13 — Tuesday 1 September

Contradictions — the subtlest capability, plus a measured latency number

WHY TODAY MATTERS

Catching a candidate contradicting themselves across turns is the most impressive thing this system does, and the hardest to get right on a free-tier model. The trick is not a better prompt — it is a smaller question.

PERSON A — MEASURE THE EXPERIENCE

Instrument end-to-end latency

WHAT Log end-of-speech to first audio byte across ten turns; report median and p90.

WHY *Report the median and p90, never the best case — the worst turns are what a judge remembers. Cross-check against Agora's own per-turn metrics from the turn-details endpoint. Almost no team will bring a measured number, and it reads as production maturity.*

PERSON B — CATCH THE CONFLICT

Narrow the comparison before writing the prompt `src/analysis/contradiction.py`

WHAT Compare each new claim only against prior claims **on the same topic**.

HOW

```
candidates = ledger.by_topic(new.topic)
prompt: "Do these conflict? Answer only
        if they cannot both be true.
        Quote both sides."
→ {"conflict": bool, "quote_a": "",
    "quote_b": "", "why": ""}
```

WHY *Handing a small model twenty unrelated claims and asking "any contradictions?" produces confident nonsense — it will find conflicts between statements about different subjects entirely. Restricting to one topic turns an open-ended reasoning problem into a narrow comparison, which is the kind of task these models do reliably. This is why `ClaimLedger.by_topic()` was built on Day 4.*

Tune for precision, not recall

WHAT Bias the prompt toward "no conflict" when uncertain, and require both quotes.

WHY *A false accusation on stage — the panel challenging a contradiction that is not there — is far more damaging than quietly missing one. Requiring both verbatim quotes also gives you the on-screen evidence that makes the catch legible to a judge.*

EVENING CHECKPOINT The harness's bait on turns 2 and 3 is caught and challenged with both quotes shown · you **Agora ~10 · total 95**
have a median latency figure

6 Week 3 — Finish and Submit (Days 14–18)

Two build days, then everything stops. The final three days are integration, assets and submission — and they consistently take longer than teams expect.

Day 14 — Wednesday 2 September

The report — what turns a demo into a product

WHY TODAY MATTERS

A live conversation impresses for four minutes. A written assessment where every judgement cites a timestamped quote is what makes a judge think this could actually be sold to a company.

PERSON A — MULTILINGUAL SUPPORT

Enable Hindi answering

WHAT Set the ASR language or enable multilingual recognition; choose TTS voices with the matching locale; add one line to each persona prompt instructing it to reply in the candidate's language.

WHY "Multilingual voice experiences" is explicitly listed by the organisers under what they want built. It costs a configuration change rather than a workstream, and for an Indian hiring context it is a genuinely useful feature rather than a demo trick.

Add a UI language toggle

WHY So it can be demonstrated deliberately on camera rather than hoped for.

PERSON B — THE ASSESSMENT

Aggregate by competency [src/analysis/report.py](#)

HOW

```
per competency:
    average score, strengths[], gaps[]
    evidence[] → (turn_id, timestamp, quote)
overall: hire / no-hire signal
RULE: no sentence without a citation
```

WHY Aggregating by competency rather than by question is what makes the output useful to a recruiter — they care whether the candidate can reason about trade-offs, not how they did on question four.

Enforce the citation rule

WHAT Every claim carries turn ID, timestamp and verbatim quote.

WHY This is PS11 capability 9 and the strongest answer to the obvious judge question, "how do we know the AI is not making this up?" Every sentence traces to something the candidate actually said.

Render to PDF with WeasyPrint

WHY You are already producing the report as HTML for the browser view, so PDF export costs almost nothing extra — and a downloadable PDF is what makes it feel like a product.

Build the scenario library [src/conductor/scenarios.py](#)

WHAT Three or four role-play scenarios per persona, entered via `launch_scenario`, floor locked to the owner until complete.

WHY PS11 capability 6. The floor lock matters: a role-play interrupted by another persona after one turn is not a scenario, it is a non sequitur.

EVENING CHECKPOINT The report contains no uncited claim · a candidate can answer in Hindi and the panel continues

Agora ~10 · total 105

Day 15 — Thursday 3 September

Human in the loop — the last feature day

WHY TODAY MATTERS

Last chance to add anything. The observer view is the final differentiator; persona tuning is the highest-value hour available. If time forces a choice, tune the personas.

PERSON A — THE OBSERVER VIEW

Build the recruiter dashboard [src/web/observer.html](#)

WHAT Live transcript, current floor holder, difficulty level, raised flags, plus three buttons: *push harder*, *move on*, *hand to Product*.

WHY *"Human-in-the-loop AI systems" is another item the organisers explicitly listed. It also answers the sharpest question a hiring-industry judge will ask — "would you let an AI make this call alone?" — with "no, and here is the supervision interface." It is half a day because the conductor already accepts directives.*

Cut this first if the day runs long

WHY *It is the only genuinely optional item left. Protecting Day 16 integration matters more than any additional feature.*

PERSON B — TUNING

Render the report in-browser with clickable citations

WHY *Clicking a citation and jumping to that transcript line is a five-second demo moment that proves the evidence chain is real rather than decorative.*

Run a full persona tuning pass

WHAT Three interviews; tighten prompts until each persona sounds genuinely distinct and stays terse.

WHY *The highest-value hour of the fortnight. On free-tier models, prompt quality affects the perceived experience more than any architectural change you could still make. Llama holds a persona well but drifts toward a generic assistant voice; short prompts with one explicit tone example fix most of it.*

EVENING CHECKPOINT A human can nudge the panel mid-interview and the next question visibly changes

Agora ~10 · total 115

Day 16 — Friday 4 September

Integration and feature freeze — both people, no new features

WHY TODAY MATTERS

Every bug found today is worth more than any feature you could add. This is also the day you verify the eleven capabilities against a live session rather than against your own code, which is where teams discover something has been quietly broken for a week.

BOTH — RUN IT FOR REAL, THEN BREAK IT DELIBERATELY

Three full interviews at `PANEL_SIZE=3`

WHAT Different topics, different candidate personalities. One plays the candidate, the other watches the logs.

WHY *Different personalities matter — a confident candidate and a hesitant one exercise completely different paths through the difficulty ladder and the vagueness detector.*

Walk the PS11 list against a live session

WHY *Against a live session, not the codebase. "The code for that exists" and "a judge would see that happen" are very different claims, and only the second one scores.*

Break it on purpose

WHAT Kill the network mid-interview. Exhaust a rate limit. Confirm the fallback chain engages and teardown still runs.

WHY *You want to discover your failure modes today, in private, rather than on stage in Delhi.*

Check for orphaned agents after every test

WHY *A survivor from a crashed session bills silently. With ~125 minutes left at this point, one forgotten agent is a real problem.*

Freeze at end of day

WHY Anything unfinished gets cut, not rushed. A smaller demo that runs beats a larger one that fails — and the rules forbid adding features after Round 4 anyway.

FEATURE FREEZE Three clean end-to-end interviews · all eleven PS11 capabilities observed live · no known crashes **Agora ~60 · total 175**

Day 17 — Saturday 5 September

Assets — the day every team underestimates

WHY TODAY MATTERS

The judges experience your project through a video, a README and a link. A brilliant system with a confusing video scores below an average system with a clear one.

PERSON A — DEPLOY AND DOCUMENT

Final deployment, tested cold

WHAT Open the link on a device that has never run the project, on a network that is not yours.

WHY Your laptop has cached credentials, permitted microphone access and local DNS. A judge's machine has none of that. This test catches the failures that only appear for strangers.

Write the README

WHAT Architecture, stack, setup — plus a **top-level heading** reading "How Agora Conversational AI is integrated".

WHY The organisers list that explanation as its own submission item. Making it a top-level heading rather than a buried paragraph means a scanning judge cannot miss it.

State the build disclosure and check licences

WHAT Confirm the project was built 20 Aug – 6 Sep, point at the commit history, and verify every model and library licence.

WHY Building from scratch makes this easy — but the rules require it explicitly, and teams are responsible for their right to use every third-party resource.

PERSON B — THE DEMO VIDEO

Record to this script, using reserved clean API keys

HOW

0:00 disclosure spoken and shown
0:20 Priya opens – distinct voice
0:45 follow-up quotes the answer
1:15 CANDIDATE INTERRUPTS
1:35 handoff fires on screen
1:50 Arjun challenges the business case
2:20 contradiction caught, both quotes
2:45 difficulty meter drops
3:00 Meera runs a role-play scenario
3:20 candidate switches to Hindi
3:35 recruiter clicks "push harder"
3:50 cited report renders
4:10 latency figures + architecture

WHY Every beat maps to a scored capability, in an order that builds. Show, do not narrate — a judge should see the interruption happen, not hear you claim it works. Budget two or three takes and use keys you have not touched all week, so a rate limit cannot ruin a good run.

EVENING CHECKPOINT Deployment opens cleanly on a stranger's device · video recorded and uploaded · README **Agora ~60 · total 235**
complete

Day 18 — Sunday 6 September

Submit — target 2 PM, not 11 PM

WHY TODAY MATTERS

The deadline is 11:00 PM IST. Every year, teams with working projects miss it through upload failures, portal timeouts and time-zone arithmetic. Treat the afternoon as the deadline.

BOTH — FINALISE, VERIFY, SUBMIT

Final deck pass

WHAT Fold in the measured latency figures and a screenshot of the cited report.

WHY *Those two artefacts are your strongest evidence of engineering maturity, and neither existed when you wrote the deck on Day 7.*

Rehearse the online presentation twice, timed

WHY *Round 4 includes a live presentation, not just an upload. An unrehearsed demo overruns and gets cut off before the best part.*

Verify every link while logged out

WHAT Repo public, deployment reachable, video playable.

WHY *The guidelines state plainly that "incomplete or inaccessible submissions may not be considered." A private repo is a scoring event, not an administrative detail.*

Submit by 2 PM, then stop

WHAT Name and description · PS11 · both members' details · repo · deployment link · video · architecture doc · Agora integration section · deck · build disclosure.

WHY *No features can be added after Round 4 in any case — 8–11 September is polish and rehearsal only. Submitting early costs nothing and removes the last remaining risk.*

DONE Submitted with nine hours to spare · roughly 55 agent-minutes still in reserve

Agora ~10 · total 245

7 Budget and Risk

7.1 The agent-minute budget

Agora bills per **agent-minute**, not per session-minute. Three personas in a ten-minute interview consume thirty agent-minutes, not ten. That makes 300 free minutes equal to roughly ten full three-persona interviews *for the entire month* — covering development, testing, rehearsal, the video and the live demo.

PERIOD	SPEND	RUNNING	NOTES
Days 1–2 — scaffold and gateway	0	0	No Agora needed yet.
Days 3–6 — smoke, silence, panel	35	35	Four-minute sessions only.
Days 7–9 — deck and tools	5	40	Nearly everything runs through the harness.
Day 10 — milestone session	25	65	Record it as fallback footage.
Days 11–15 — daily checks	50	115	<code>PANEL_SIZE=2</code> ; saves a third.
Day 16 — integration	60	175	Three full three-persona interviews.
Day 17 — video takes	60	235	Two or three takes at full panel size.
Day 18 — reserve	10	245	~55 minutes unspent.

IF THE BUDGET STARTS SLIPPING

In this order: drop to `PANEL_SIZE=2` for all testing · cut test sessions to three minutes · move anything still being debugged back onto the mock harness · ask on Discord again. Do **not** economise on Day 16 integration or Day 17 video takes — those two days are what actually gets scored.

7.2 Risks worth naming

RISK	IMPACT	MITIGATION
Agora interrupt test fails on Day 3	Fatal	Escalate on Discord the same day. B's lane continues regardless via the mock harness, so only half the team is blocked.
Empty completions do not silence an agent	High	Tested Day 4. Fallback: <code>remote_rtc_uids: []</code> plus subscription swaps via <code>update</code> .
Free agent-minutes exhausted	Fatal	Ask Discord Day 1 · mock harness from Day 5 · <code>PANEL_SIZE</code> flag · guaranteed teardown.
Rate limit during the live demo	High	Provider fallback chain (Day 9) · reserved untouched keys · separate dev keys per person.
Contradiction detector produces false positives	Medium	Topic-narrowed comparison, both quotes required, tuned for precision (Day 13).
Personas sound identical	Medium	Distinct voices chosen Day 3 · dedicated tuning pass Day 15.
Missing registration or deck deadline	Fatal	Day 6 and Day 9 respectively, both flagged in the plan.
Live demo fails on venue Wi-Fi	High	Day 10's recorded session is your fallback footage; rehearse on a hotspot before the finale.

7.3 What to cut, in order

If you fall behind, cut from the bottom of this list first. Everything above the line is either a scored PS11 capability or a submission requirement.

1. **Observer view (Day 15)** — a bonus capability, not a PS11 requirement.

2. **Multilingual (Day 14)** — same; nice to have, explicitly listed but not scored under PS11.
3. **Scenario library (Day 14)** — reduce from four scenarios per persona to one.
4. **Third persona** — drop to two. PS11 says "one or more", and it saves a third of your agent-minutes.
5. **Everything else is load-bearing.** Below this point you start losing scored capabilities.

WHERE THE SLACK IS

This plan carries roughly four spare person-days across eighteen days. That slack is deliberate and it is not free time — it absorbs the day Agora behaves unexpectedly, the day a rate limit blocks you, and the day something simply takes twice as long as it should. If you reach Day 16 with it unused, spend it on persona tuning. Nothing else improves the demo as much per hour.